

Statistics 651

Markov Chains

Winter 2026

Objectives

1. Construct Markov chains and find their stationary distributions
2. Understand the conditions under which convergence to the stationary distribution is guaranteed
3. Sample from Markov chains to verify their long-range behavior

Resources

- BDA3: Ch 11 intro
- BIDA: Ch 6.3.1
- [Robert and Casella \(2004\)](#): Ch 6 (Ch 4.6 in 1999 edition available online through the [BYU Library](#))
- [Tierney \(1994\)](#)

Motivation

Bayesian inference is built on joint distributions

$$p(\boldsymbol{x}, \boldsymbol{\theta}) = f(\boldsymbol{x} \mid \boldsymbol{\theta}) \pi(\boldsymbol{\theta}) .$$

The posterior distribution $\pi(\boldsymbol{\theta} \mid \boldsymbol{x})$ requires integration of dimension equal to that of $\boldsymbol{\theta}$.

How do the integration techniques from earlier scale in dimensions?

Stochastic processes

Definition

A **stochastic process** is a collection of

- random variables $X(t) \in \mathcal{X}$ and
- associated index $t \in \mathcal{T}$,

such that for any value of t , $X(t)$ has a probability distribution.

For any finite collection $\{t_1, \dots, t_n\}$, there exists a joint distribution of $(X(t_1), \dots, X(t_n))$, called the **finite-dimensional distributions** of the process.

Goal of MCMC:

Generate sequences $X(t)$ for $t = 1, 2, \dots$ that are easy to construct, and have the property that for any given t , $X(t)$ comes from the target distribution.

Joint distributions with the chain rule

Chain rule for distributions

Any joint distribution for $(X_1, X_2, \dots, X_n)$ can be constructed sequentially:

$$p(x_1, x_2, \dots, x_n) = p_1(x_1) p_2(x_2 \mid x_1) p_3(x_3 \mid x_2, x_1) \cdots p_n(x_n \mid x_{n-1}, \dots, x_1)$$

Markov property

Definition

Consider a discrete-time stochastic process $X_t := X(t)$ for $t = 0, 1, 2, \dots$

The process has the **Markov property** if the distribution of X_t , given the entire past X_{t-1}, X_{t-2} , etc., is conditionally independent of all previous values except the most recent.

That is,

$$p_t(x_t \mid x_{t-1}, x_{t-2}, \dots) = p_t(x_t \mid x_{t-1}) \text{ for all } t.$$

The Markov property simplifies construction of the joint distribution:

$$p(x_0, x_1, \dots, x_n) = p_0(x_0) \times \prod_{t=1}^n p_t(x_t \mid x_{t-1})$$

Markov chains

Definition

A discrete-time **Markov chain** is a stochastic process for $\{X_t : t = 0, 1, \dots\}$ with the Markov property.

It is characterized by

- an initial value x_0 or distribution $p_0(x_0)$, and
- the set of transition distributions $p_t(x_t \mid x_{t-1})$ for $t = 1, 2, \dots$

Note:

We will assume homogeneity of the transition distribution, that is,
 $p_t(x_t \mid x_{t-1}) = p(x_t \mid x_{t-1})$ for all t .

Markov chains

Example of discrete-time chain on discrete space

Random walk on a ring.

State space $\mathcal{X} = \{1, 2, 3, 4, 5\}$

Transition distributions:

$$P(X_{t+1} = 3 | X_t = 2) = .5 \quad \text{one prob from that dist}$$

Markov chains

Definition

If the transition distributions of a Markov chain with discrete state space $\mathcal{X}$ are unchanging (independent of t), then they can be collected into a **transition probability matrix**, $\mathbf{P}$.

Note: We follow the convention that rows index the *from* states and columns index the *to* states.

Example

Write the transition matrix for the random walk on a ring.

$$\mathbf{P} = \begin{matrix} & \begin{matrix} \text{to} \\ 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} \text{from} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{pmatrix} 0 & .5 & 0 & 0 & .5 \\ .5 & 0 & .5 & 0 & 0 \\ 0 & .5 & 0 & .5 & 0 \\ 0 & 0 & .5 & 0 & .5 \\ .5 & 0 & 0 & .5 & 0 \end{pmatrix} \end{matrix}$$

Markov chains

💡 Example of discrete-time chain on continuous space

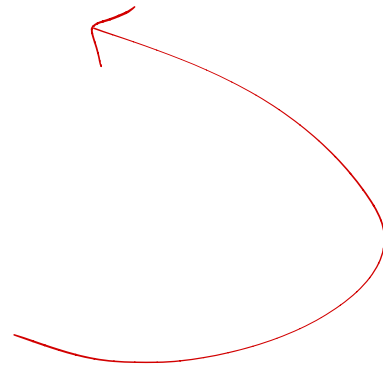
Autoregressive model:

$$X_0 = 0$$

$$X_t \mid X_{t-1} \stackrel{\text{ind}}{\sim} \text{Normal}(\phi X_{t-1}, 1) \text{ for } \phi \in (\text{~~0~~, 1).$$

State space $\mathcal{X} = \mathbb{R}$

Transition distributions:



Forecasting probabilities

💡 Example (discrete space)

Consider a discrete Markov chain with two states $\mathcal{X} = \{0, 1\}$.

Let $X_0 \sim \text{Bern}(0.8)$,

$X_t \mid (X_{t-1} = 0) \sim \text{Bern}(0.5)$,

$X_t \mid (X_{t-1} = 1) \sim \text{Bern}(0.3)$, for all $t = 1, 2, \dots$

$$P = \begin{matrix} & \begin{matrix} 0 & 1 \end{matrix} \\ \begin{matrix} 0 \\ 1 \end{matrix} & \begin{pmatrix} .5 & .5 \\ .7 & .3 \end{pmatrix} \end{matrix}$$

Find the following

a. The joint PMF of X_1 and X_2

$$\begin{matrix} & \begin{matrix} 0 & 1 \end{matrix} \\ \begin{matrix} 0 \\ 1 \end{matrix} & \begin{pmatrix} .2(.5)(.5) + .8(.7)(.5) & .2(.5)(.5) + .8(.7)(.5) \\ .2(.5)(.7) + .8(.3)(.7) & .2(.5)(.3) + .8(.3)(.3) \end{pmatrix} \end{matrix}$$

b. The marginal PMF of X_2 $\sum_{x_1} P(x_1, x_2)$

c. The conditional PMF of $X_2 \mid X_1$ P

d. The conditional PMF of $X_3 \mid X_1$ $\sum_{x_2} P(x_2 \mid x_1) P(x_3 \mid x_2) = P^2$

e. The conditional PMF of $X_{t+n} \mid X_t$ for $n \geq 1$ P^n

f. The marginal PMF of X_n for $n \geq 1$ $\sum_{x_0, \dots, x_{n-1}} p(x_0, x_1, \dots, x_n) = (.2, .8) P^n$

Long-term behavior

Example (discrete space)

$$\mathbb{P}(X_{t+n} = i \mid X_t = j) = P_{ij}^n$$

What happens as n gets large?

every row converges to the same distrib $\underline{\pi} \approx \begin{pmatrix} .58333 \\ .41667 \end{pmatrix}$

Forecasting probabilities

Example (continuous space)

Autoregressive model:

$$X_0 = 0$$

$$X_t \mid X_{t-1} \stackrel{\text{ind}}{\sim} \text{Normal}(\phi X_{t-1}, 1) \text{ for } \phi \in (0, 1).$$

Find the following

- The marginal distribution of X_1 $N(0, 1)$
- The conditional distribution of $X_2 \mid X_1$ $N(\phi X_1, 1)$
- The conditional distribution of $X_3 \mid X_1$ $\int N(x_3 \mid \phi x_2, 1) N(x_2 \mid \phi x_1, 1) dx_2$
- The conditional distribution of $X_{t+n} \mid X_t$ for $n \geq 1$

can also write
 $X_3 = \phi X_2 + \varepsilon_3$ where $\varepsilon_3 \sim N(0, 1)$
 $= \phi(\phi X_1 + \varepsilon_2) + \varepsilon_3$
 $= \phi^2 X_1 + \phi \varepsilon_2 + \varepsilon_3$
 which is distributed Normal
 with mean $\phi^2 X_1$
 and variance $\phi^2 + 1$
 (since $\varepsilon_2, \varepsilon_3$ indep)

$$X_{t+n} = \phi^n X_t + \phi^{n-1} \varepsilon_{t+1} + \phi^{n-2} \varepsilon_{t+2} + \dots + \phi \varepsilon_{t+n-1} + \varepsilon_{t+n}$$

which is Normal w mean $\phi^n X_t$ and variance $1 + \phi^2 + \phi^4 + \dots + (\phi^{n-1})^2$

Long-term behavior

Example (continuous space)

What happens as $n \rightarrow \infty$?

$$X_n \sim N(0, V)$$

$$\text{where } V = 1 + \phi^2 + \phi^4 + \dots$$

$$= \sum_{i=0}^{\infty} (\phi^2)^i$$

$$= \frac{1}{1 - \phi^2}$$

Simulate chains

💡 Example (continuous space)

Verify this long-term behavior by simulating long chains in each of the discrete and continuous cases.

$$P = \begin{pmatrix} .5 & .5 \\ .7 & .3 \end{pmatrix}$$

prob distrib of x_{t+1} if $x_t = 0$

prob distrib of x_{t+1} if $x_t = 1$

Stationary distribution

Definition

Let $K(x, B)$ denote the transition probability of going from $x \in \mathcal{X}$ to set $B \in \mathcal{B}(\mathcal{X})$.

A probability distribution Π that satisfies

$$\mathbb{P}_{\Pi}(B) = \int K(x, B) \Pi(\mathrm{d}x)$$

is called a **stationary distribution** of the Markov chain.

If $\mathcal{X}$ is discrete, then the definition can be written as $\boldsymbol{\pi}^{\top} \boldsymbol{P} = \boldsymbol{\pi}^{\top}$.

Stationary distribution

How does this help us?

- If we start with a sample from the stationary distribution, and run the chain, then the draws will all marginally come from the stationary distribution (though they are dependent).
- We showed earlier that the target distribution is the stationary distribution of a Gibbs sampler.

What else do we need?

Convergence

Definition

A Markov chain with stationary distribution Π is **ergodic** if for every starting distribution Π_0 ,

$$\lim_{n \rightarrow \infty} \left\| \int K^n(x, B) \Pi_0(dx) - \mathbb{P}_\Pi(B) \right\| = 0$$

for every $B \in \mathcal{B}(\mathcal{X})$, where $K^n(x, B)$ denotes n applications of the transition distribution.

Furthermore, this distance is decreasing in n .

See Ch 6.6 of [Robert and Casella \(2004\)](#) (Ch 4.6 in 1999 edition available online through the [BYU Library](#)).

Convergence

Conditions for convergence

A Markov chain is ergodic and converges to a unique stationary distribution Π if it has all of the following properties:

- **Aperiodic**
 - Return times to $B \in \mathcal{B}(\mathcal{X})$ with positive probability have GCD equal to one.
- **Irreducible**
 - For all $x \in \mathcal{X}$ and $B \in \mathcal{B}(\mathcal{X})$ with $\mathbb{P}_{\Pi}(B) > 0$, there exists n such that $K^n(x, B) > 0$.
 - “You can get anywhere from anywhere, *eventually*.”
- **Harris recurrent**
 - With probability one, any $B \in \mathcal{B}(\mathcal{X})$ with $\mathbb{P}_{\Pi}(B) > 0$ will be visited infinitely often.

Sufficient conditions to check (BIDA p. 150):

$\mathbb{P}_{\Pi}(B) > 0$ if and only if $K(x, B) > 0$ for all $x \in \mathcal{X}$ and $B \in \mathcal{B}(\mathcal{X})$.

Furthermore, we only need some $n > 0$ such that this holds for K^n

Conditions for convergence

Examples

Check the sufficient conditions for ergodicity in each example:

a. Random walk on a ring

There exists $n > 0$ such that P^n has all non-zero entries

b. Bernoulli chain

$\begin{pmatrix} .8 & .5 \\ .7 & .3 \end{pmatrix}$ all non-zero

c. Autoregressive model

Transition distrib has pos. dens on $\mathbb{R}$

Building an MCMC algorithm

Markov chain Monte Carlo (MCMC) algorithms specify transition distributions that converge to the target distribution Π .

To (theoretically) guarantee convergence, establish

1. Ergodicity

2. Π -invariance: $\int K(x, B) \Pi(\mathrm{d}x) = \mathbb{P}_{\Pi}(B)$

- A sufficient (but not necessary) condition for Π -invariance is that Π and K satisfy the **detailed balance equations**, that is,

$$\mathbb{P}_{\Pi}(A) \int_A K(x, B) \Pi(\mathrm{d}x) = \mathbb{P}_{\Pi}(B) \int_B K(x, A) \Pi(\mathrm{d}x)$$

Common transition algorithms

Each of the popular algorithms require that the target distribution is **known only up to proportionality(!!)**

We explore:

- Gibbs sampling
- Metropolis-Hastings
- Slice sampling

as well as hybrid combinations.

Speed of convergence

Spectral properties

Consider an ergodic Markov chain with discrete sample space, $K \times K$ transition matrix $\mathbf{P}$, and stationary distribution $\boldsymbol{\pi}$. Then,

1. $\boldsymbol{\pi}$ is an eigenvector of $\mathbf{P}$ with eigenvalue 1.
2. All other eigenvalues are between -1 and 1.

Consider the spectral decomposition of $\mathbf{P}^\top = \mathbf{Q}\mathbf{D}\mathbf{Q}^{-1}$ with $\mathbf{D}$ diagonal.

Then, $(\mathbf{P}^\top)^n = \mathbf{Q}\mathbf{D}^n\mathbf{Q}^{-1}$

 contains eigenvalues

 The eigenvalue of 1 will persist for all n
All other eigenvals $\rightarrow 0$

So all that is left are eigenval of 1 and $\boldsymbol{\pi}$, the stationary distribution.

How quickly this occurs depends on the magnitudes
of the other eigenvalues.